

Please follow the structure, highlighting guidelines, and Do's/Don'ts below for your final Software Architecture and Handover Document (SAD).

1. Suggested Final Documentation Structure

Your final document must be cohesive, combining the deliverables from all Sprints into a single master file. Use the Infosys official Word templates and adhere to the following hierarchy:

- **I. Title & Meta Information:**
 - Project Title: NeoBank 360 – Unified Digital Banking Platform.
 - Program: PMIS (Pradhan Mantri Internship Scheme) – Infosys Limited.
 - Location: Bhubaneswar DC | Lab 4 & Lab 5.
 - Team Members & Module Ownership: Clearly state who built what.
- **II. Executive Summary:** A high-level overview of the NeoBank platform, the problems it solves, and the business value of the modules you developed.
- **III. System Architecture & Tech Stack:**
 - Provide the tech stack rationale (Why Angular? Why Spring Boot? Why MySQL?).
 - Include a Component Interaction Diagram illustrating data flow from the UI to the database.
 - Provide the final ER Diagram mapping all 11 database tables, including foreign keys and the `system_audit_log`.
- **IV. Core Modules Breakdown (FR-1 to FR-8):** Summarize the logic for Authentication, Accounts, Budgets, Bills, Loans, and Analytics without getting bogged down in minor details.
- **V. Security & Authentication Posture:** Explain your implementation of stateless JWT tokens, BCrypt hashing, and Spring Security Role-Based Access Control (RBAC).
- **VI. Performance & Enhancements:** Document how you eliminated N+1 query issues, implemented composite MySQL indexes, and utilized `@Cacheable`.
- **VII. Deployment & Infrastructure Guide:** Step-by-step instructions for running the application using Docker, including the required environment variables (e.g., `application-prod.yml`).
- **VIII. Quality Assurance Summary:** Summarize the results from your E2E tests and JMeter/Postman load testing.
- **IX. Challenges & Key Learnings:** Reflect on specific technical hurdles you overcame during this internship.

2. What to Highlight vs. What to Keep Brief

What to Highlight (Focus Here):

- **The "Why" Behind Technical Decisions:** Don't just tell me you used an Interceptor. Explain *why* an AOP interceptor was the best choice for the System Audit Log without blocking the main API thread.
- **Complex Business Logic:** Break down your mathematical approaches, such as the exact reducing balance formula you used for EMI calculations and how you handled decimal rounding.
- **Performance Triumphs:** Highlight database query optimizations and caching strategies that drastically improved load times for the Admin Dashboards.
- **Foolproof Deployment:** Your Docker instructions and cloud deployment steps must be explicitly clear so anyone can spin up the project.

What to Keep Brief (Do Not Clutter):

- **Code Dumps:** Do not paste 500 lines of Java or TypeScript into a Word document. Code belongs in Git. You may use 5-10 line snippets *only* if illustrating a highly specific piece of logic.

- **Exhaustive API Definitions:** Do not list out all 40+ endpoints with their request/response bodies. Simply summarize the RESTful structure and provide a link to your exported openapi-final.json or live Swagger UI.
- **Basic Setup Tutorials:** Keep Node.js or Java JDK installation steps out of this document. Assume the reader is a senior engineer.

3. Do's and Don'ts

The Do's:

- **Do** use proper, professional English and leverage spelling/grammar tools before submission.
- **Do** use clean, high-resolution diagrams (Draw.io, Lucidchart, etc.) instead of messy whiteboard photos.
- **Do** credit your peers. Software engineering is a team sport; ensure module ownership is accurately represented.
- **Do** ensure all formatting, fonts, and heading sizes align with Infosys enterprise standards.

The Don'ts:

- **Don't** leave placeholder text (e.g., [Insert Diagram Here]) in your final submission.
- **Don't** expose sensitive data. Ensure no real passwords, active JWT tokens, or production database credentials are typed into this document or visible in your screenshots.
- **Don't** submit a 100-page document full of fluff. Keep it concise, technical, and impactful.

Below is the mandated structure for your presentation, guidelines on what to focus on, and a list of Do's and Don'ts for the live demo.

1. Suggested Final PPT Structure (14-Slide Minimum)

Your slide deck must tell a cohesive story, moving from the business problem to the technical solution, culminating in the live demo.

- **Slide 1: Title Slide:** NeoBank 360, Team Name/Members, Date, Infosys PMIS Program.
- **Slides 2-3: Problem Statement & Solution:** What fragmented banking issue does NeoBank solve, and how does our unified platform address it?
- **Slide 4: Technology Stack:** High-level overview of Angular, Spring Boot, MySQL, JWT, and Docker.
- **Slide 5: System Architecture:** Your best, highest-resolution component interaction diagram from your SAD.
- **Slides 6-8: Key Features Walkthrough:** High-level summaries of Core Banking, Budgeting, and the Loan Approval Workflow.
- **Slides 9-10: Advanced Analytics:** Screenshots of the Admin System Logs and User Spending Doughnut/Area charts.
- **Slide 11: Execution Timeline:** A visual roadmap of how we accomplished this across the 60-day, 5-sprint lifecycle.
- **Slide 12: Intern Contributions:** A clear matrix of module ownership and individual achievements.
- **Slide 13: Challenges & Learnings:** Key technical roadblocks (e.g., EMI math, N+1 query optimization) and how the team overcame them.
- **Slide 14: Live Demo Notice:** A transition slide signaling the shift to screen share for the live application walkthrough.

2. What to Highlight vs. What to Keep Brief

What to Highlight (Focus Here):

- **High-Quality Visuals:** Use architecture diagrams and clean, high-resolution screenshots of the UI (especially the analytics charts).
- **Business Impact:** Connect your technical features to real-world value. Explain *why* the automated EMI schedule generation saves administrative time.
- **Your Specific Contributions:** Be ready to speak confidently about the exact modules you built and optimized.

What to Keep Brief (Do Not Clutter):

- **Walls of Text:** Nobody reads paragraphs on a slide. Use short, punchy bullet points. You should be *speaking* the details, not reading them off the screen.
- **Deep Code Dives:** Do not paste code blocks into the presentation unless it is a very brief (3-4 lines) example of something highly innovative. Save deep technical explanations for the Q&A session.

3. Do's and Don'ts

The Do's:

- **Do** practice your speaker transitions. Handing off the presentation from one intern to the next should feel seamless and rehearsed.
- **Do** prepare fallback screenshots. Live demos can break due to network issues. Have a folder of screenshots ready for every step of your demo script just in case.
- **Do** use the official Infosys PowerPoint template. Maintain consistent fonts, branding colors, and spacing throughout.

- **Do** seed your database heavily before the demo. An analytics dashboard with two transactions looks unimpressive. Seed thousands of records so the charts look enterprise-ready.

The Don'ts:

- **Don't** speak over each other during the demo. Stick strictly to your assigned speaking roles.
- **Don't** rush. Speak clearly and confidently. Pause to let the audience digest complex architecture diagrams.
- **Don't** panic during the Q&A. If you are asked a technical question you don't know the answer to, it is okay to say, "I focused primarily on the frontend routing for that module, but my colleague [Name] handled the backend logic and can answer that."

4. Prompt for Interns to Generate the PPT Content

You can provide this prompt to your team so they can generate the structured outline and speaker notes for their slides. They must then move this content into the official Infosys PowerPoint template.

Prompt for Interns:

"Act as a Senior Technical Presenter for Infosys Limited. I need to create a 14-slide professional presentation deck for my team's PMIS internship project: 'NeoBank 360 – Unified Digital Banking Platform.'

Please generate a slide-by-slide outline including:

1. The Slide Title.
2. 3 to 4 concise, impactful bullet points for the slide body (suitable for an enterprise audience).
3. Detailed 'Speaker Notes' for each slide to guide what the presenter should actually say.

Structure the 14 slides exactly as follows:

1. Title Slide
2. The Banking Problem
3. The NeoBank Solution
4. Technology Stack
5. System Architecture
6. Core Banking & Transactions
7. Budgeting & Bill Management
8. Loan Application & EMI Engine
9. Advanced Admin Analytics & System Logs
10. Advanced User Wealth Analytics
11. Entire Sprint Day Execution Timeline
12. Team Contributions & Module Ownership
13. Technical Challenges & Learnings
14. Transition to Live Demo